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# ASSIGNMENT - CONFIGURING A SECURE MAIL SYSTEM WITH FREEIPA, POSTFIX, DOVECOT, AND EVOLUTION

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## PART 0 – DOWNLOAD AND IMPORT VM

1. Using the flowing link, download and import AlmaLinux VM and import twice, once as **AlmaLinux-Server** and the other VM **AlmaLinux-Client**. (Username **atohme/alma**)

[Linux-VM](#)

## PART I - FREEIPA SERVER INSTALLATION AND cONFIGURATION

1. Log in **AlmaLinux-Server** graphically and install FreeIPA Server with DNS support.

```
atohme@localhost:~$ sudo su -
[sudo] password for atohme:
root@localhost:~# dnf install ipa-server ipa-server-dns postfix telnet dovecot -y
AlmaLinux 9 - AppStream                                42 kB/s | 4.2 kB | 00:00
AlmaLinux 9 - AppStream                                39 MB/s | 13 MB | 00:00
AlmaLinux 9 - BaseOS                                    47 kB/s | 3.8 kB | 00:00
AlmaLinux 9 - BaseOS                                    40 MB/s | 9.9 MB | 00:00
AlmaLinux 9 - Extras                                    44 kB/s | 3.8 kB | 00:00
Dependencies resolved.
234 kB/s | 22 kB | 00:00

Package Architecture Version Repository Size
Installing:
dovecot x86_64 1:2.3.16-18.el9_8 appstream 4.7 M
ipa-server x86_64 4.13.1-3.el9_8.2 appstream 407 k
ipa-server-dns noarch 4.13.1-3.el9_8.2 appstream 54 k
postfix x86_64 2:3.5.25-2.el9 appstream 1.4 M
telnet x86_64 1:0.17-85.el9 appstream 63 k
```

```

perl-threads-1:2.25-406.el9.x86_64
pki-jackson-annotations-2.19.1-1.el9_6.noarch
pki-jackson-databind-2.19.1-1.el9_6.noarch
pki-jackson-jaxrs-providers-2.19.1-1.el9_6.noarch
pki-resteasy-client-3.0.26-19.el9.noarch
pki-resteasy-jackson2-provider-3.0.26-19.el9.noarch
postfix-2:3.5.25-2.el9.x86_64
python3-argcomplete-1.12.0-5.el9.noarch
python3-babel-2.9.1-2.el9.noarch
python3-cffi-1.14.5-5.el9.x86_64
python3-decorator-4.4.2-6.el9.noarch
python3-gssapi-1.6.9-5.el9.x86_64
python3-ipaclient-4.13.1-3.el9_8.2.noarch
python3-ipaserver-4.13.1-3.el9_8.2.noarch
python3-jcryption-1.5.0-3.el9_8.noarch
python3-ldap-3.4.3-2.el9.x86_64
python3-libipa-hbac-2.9.0-4.el9_8.x86_64
python3-mod_wsgi-4.7.1-12.el9_8.x86_64
python3-netifaces-0.10.6-15.el9.x86_64
python3-pyasn1-0.4.8-7.el9_7.noarch
python3-pyca-2.20-6.el9.noarch
python3-pyusb-1.0.2-13.el9.noarch
python3-sss-2.9.8-4.el9_8.x86_64
python3-sssdconfig-2.9.8-4.el9_8.noarch
rpcbind-1.2.6-7.el9.x86_64
sif4j-1.7.30-16.el9.noarch
softhsm-2.6.1-11.el9.x86_64
sssd-dbus-2.9.8-4.el9_8.x86_64
sssd-nfs-idmap-2.9.8-4.el9_8.x86_64
sssd-tools-2.9.8-4.el9_8.x86_64
tomcat-1:9.0.110-3.el9_8.noarch
tomcat-jsp-api-1:9.0.110-3.el9_8.noarch
tomcat-servlet-4.0-api-1:9.0.110-3.el9_8.noarch
t2data-java-2026b-1.el9.noarch
xnl-commons-resolver-1.2.42.el9.noarch

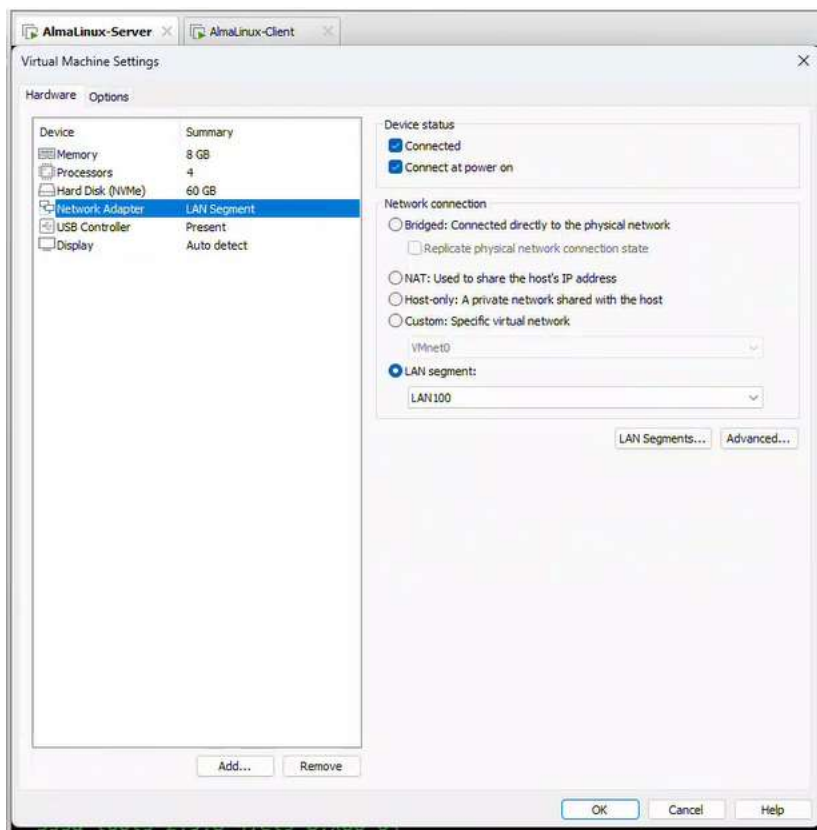
perl-threads-shared-1.01-406.el9.x86_64
pki-jackson-core-2.19.1-1.el9_6.noarch
pki-jackson-jaxrs-json-provider-2.19.1-1.el9_6.noarch
pki-jackson-module-jaxb-annotations-2.19.1-1.el9_8.noarch
pki-resteasy-core-3.0.26-19.el9.noarch
pki-resteasy-servlet-initializer-3.0.26-19.el9.noarch
publicsuffix-list-20210518-3.el9.noarch
python3-supeas-8.5.0-25.el9.noarch
python3-bind-32:9.16.23-40.el9_8.1.noarch
python3-cryptography-36.0.1-5.el9_8.x86_64
python3-dns-2.6.1-3.el9.noarch
python3-idm-pki-11.7.1-1.el9.noarch
python3-ipalib-4.13.1-3.el9_8.2.noarch
python3-jinja2-2.11.3-0.el9_5.noarch
python3-kdcproxy-1.1.0-1.el9.noarch
python3-lib389-2.6.0-6.el9_8.noarch
python3-markupsafe-1.1.1-12.el9.x86_64
python3-netaddr-0.10.1-3.el9.noarch
python3-ply-3.11-14.el9.noarch
python3-pyasn1-modules-0.4.0-7.el9_7.noarch
python3-pytr-2021.1-5.el9.noarch
python3-qrcode-core-6.1-12.el9.noarch
python3-sss-murmur-2.9.8-4.el9_8.x86_64
python3-yubico-1.3.3-7.el9.noarch
slapd-nis-0.66.0-5.el9_3.x86_64
sif4j-jdk14-1.7.30-16.el9.noarch
sqlite-3.34.1-10.el9_8.x86_64
sssd-idp-2.9.8-4.el9_8.x86_64
sssd-passkey-2.9.8-4.el9_8.x86_64
telnet-1:6.17-05.el9.x86_64
tomcat-el-3.0-api-1:9.0.110-3.el9_8.noarch
tomcat-lib-1:9.0.110-3.el9_8.noarch
ttwafdir-3.0.9-65.el9.x86_64
xnl-commons-apis-1.4.01-39.el9.noarch
xorg-x11-fonts-Type1-7.5-33.el9.noarch

Complete!
[root@localhost ~]#

```

2. Modify in the VM NAT to LAN and configure static IP, hostname, and `/etc/hosts` file.

## VM LAN



static IP, hostname

```
AlmaLinux-Server AlmaLinux-Client
Jun 15 19:42
root@localhost:~
[root@localhost ~]# nmcli con del ethernet-ens160
Connection 'ethernet-ens160' (3e8d0b36-0b77-446e-bd76-e7eafae89403) successfully deleted.
[root@localhost ~]# nmcli con add type ethernet ifname ens160 ipv4.method manual con-name LAN ipv4.addresses 192.168.15.10/24 ipv4.gateway 192.168.15.20 ipv4.dns 192.168.15.10
Connection 'LAN' (4dcb00e-4bef-4e4e-a115-3ba133da5353) successfully added.
[root@localhost ~]# hostnamectl set-hostname ipa.itmt15.local
```

```
AlmaLinux-Server AlmaLinux-Client
Jun 15 19:43
root@localhost:~
[root@localhost ~]# ip address
1: lo: <LOOPBACK,UP,LOWER UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:bc:8a:88 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.15.10/24 brd 192.168.15.255 scope global noprefixroute ens160
        valid lft forever preferred_lft forever
    inet6 fe80::a775:42fa:4575:8d94/64 scope link noprefixroute
        valid lft forever preferred_lft forever
[root@localhost ~]# hostname
ipa.itmt15.local
[root@localhost ~]#
```

## /etc/hosts

```
AlmaLinux-Server AlmaLinux-Client
Jun 15 19:46
root@localhost:~
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.15.10 ipa.itmt15.local ipa mail.itmt15.local mail
```

3. Open firewall ports for LDAP, DNS, Kerberos, HTTP/HTTPS.

```
AlmaLinux-Server AlmaLinux-Client
Jun 15 19:51
root@localhost:~
[root@localhost ~]# firewall-cmd --add-port={80,443,389,636,88,464,53}/tcp --permanent --zone=public
success
[root@localhost ~]# firewall-cmd --add-port={80,464,123,53}/udp --permanent --zone=public
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]#
```

4. Setup IPA with DNS.

```
AlmaLinux-Server AlmaLinux-Client
Jun 15 19:52
root@localhost:~
[root@localhost ~]# ipa-server-install --setup-dns

The log file for this installation can be found in /var/log/ipaserver-install.log
=====
This program will set up the IPA Server.
Version 4.13.1
```

```
AlmaLinux-Server x AlmaLinux-Client x
Activities Terminal Jun 15 20:05
root@localhost:~

Configured /etc/sss/sss.conf
Systemwide CA database updated.
Adding SSH public key from /etc/ssh/ssh_host_ed25519_key.pub
Adding SSH public key from /etc/ssh/ssh_host_ecdsa_key.pub
Adding SSH public key from /etc/ssh/ssh_host_rsa_key.pub
SSSD enabled
Configured /etc/openldap/ldap.conf
Configured /etc/ssh/ssh_config
Configured /etc/ssh/sshd_config.d/04-ipa.conf
Configuring itmt15.local as NIS domain.
Client configuration complete.
The ipa-client-install command was successful

Enabling and restarting the IPA service
=====
Setup complete

Next steps:
1. You must make sure these network ports are open:
   TCP Ports:
   * 80, 443: HTTP/HTTPS
   * 389, 636: LDAP/LDAPS
   * 88, 464: kerberos
   * 53: bind
   UDP Ports:
   * 88, 464: kerberos
   * 53: bind
   * 123: ntp

2. You can now obtain a kerberos ticket using the command: 'kinit admin'
   This ticket will allow you to use the IPA tools (e.g., ipa user-add)
   and the web user interface.

Be sure to back up the CA certificates stored in /root/cacert.p12
These files are required to create replicas. The password for these
files is the Directory Manager password
The ipa-server-install command was successful
[root@localhost ~]#
```

5. Verify IPA services and DNS configuration.

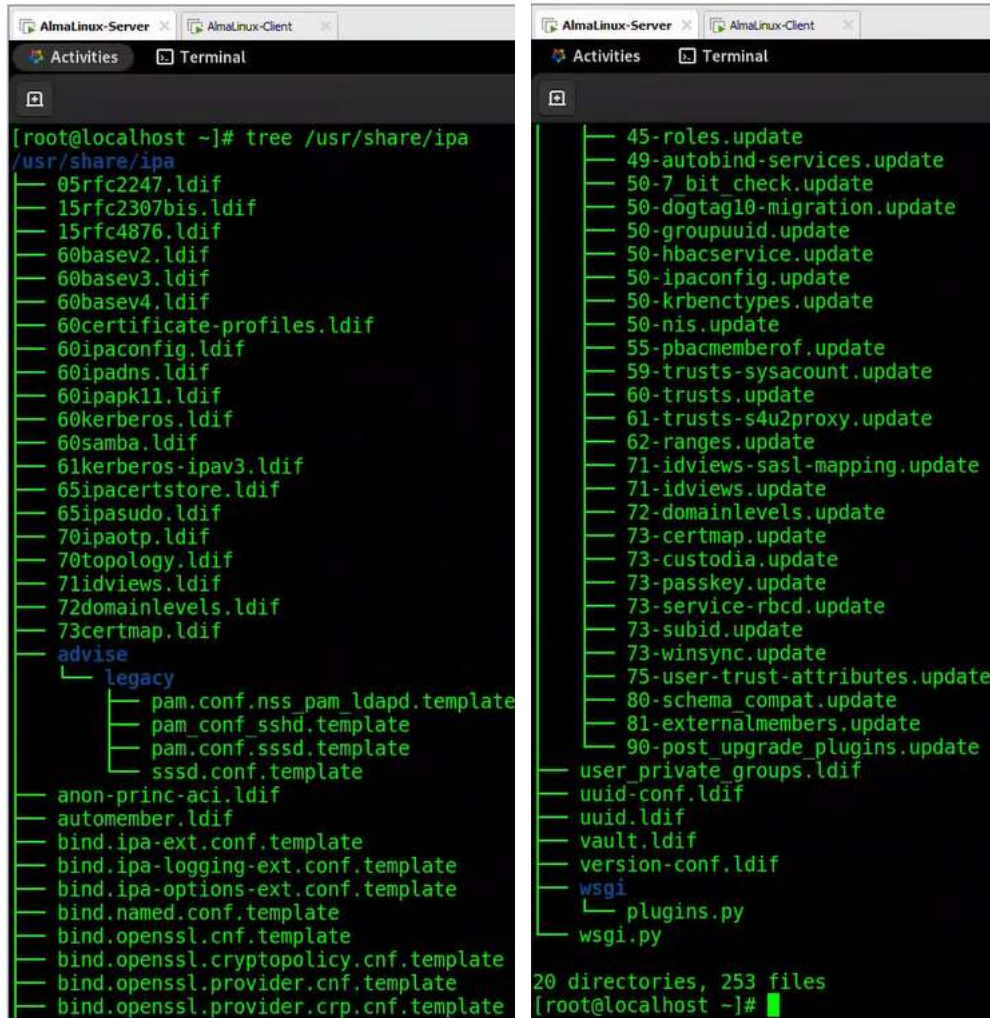
### Config files:

```
AlmaLinux-Server x AlmaLinux-Client x
Activities Terminal
[root@localhost ~]# tree /etc/ipa/
/etc/ipa/
├── ca.crt
├── custodia
│   ├── custodia.conf
│   └── server.keys
├── default.conf
├── dnssvc
│   ├── ipa-dnskeysyncd.keytab
│   ├── openssl.cnf
│   ├── softhsm2.conf
│   └── softhsm_pin_so
├── html
│   ├── ssbrowser.html
│   └── unauthorized.html
├── kdcproxy
│   ├── ipa-kdc-proxy.conf
│   └── kdcproxy.conf
├── nssdb
│   ├── cert9.db
│   ├── key4.db
│   ├── pkcs11.txt
│   └── pwdfile.txt
└── ...

5 directories, 16 files
[root@localhost ~]#
```

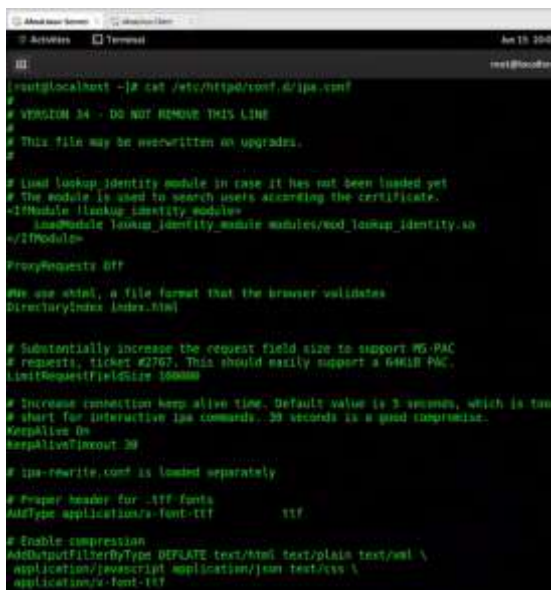


## Data files:



```
[root@localhost ~]# tree /usr/share/ipa
/usr/share/ipa
├── 05rfc2247.ldif
├── 15rfc2307bis.ldif
├── 15rfc4876.ldif
├── 60basev2.ldif
├── 60basev3.ldif
├── 60basev4.ldif
├── 60certificate-profiles.ldif
├── 60ipaconfig.ldif
├── 60ipadns.ldif
├── 60ipapk11.ldif
├── 60kerberos.ldif
├── 60samba.ldif
├── 61kerberos-ipav3.ldif
├── 65ipacertstore.ldif
├── 65ipasudo.ldif
├── 70ipaotp.ldif
├── 70topology.ldif
├── 71idviews.ldif
├── 72domainlevels.ldif
├── 73certmap.ldif
├── advise
│   └── legacy
│       ├── pam.conf.nss pam ldapd.template
│       ├── pam.conf.sshd.template
│       ├── pam.conf.sssd.template
│       └── sssd.conf.template
├── anon-princ.aci.ldif
├── automember.ldif
├── bind.ipa-ext.conf.template
├── bind.ipa-logging-ext.conf.template
├── bind.ipa-options-ext.conf.template
├── bind.named.conf.template
├── bind.openssl.cnf.template
├── bind.openssl.cryptopolicy.cnf.template
├── bind.openssl.provider.cnf.template
├── bind.openssl.provider.crp.cnf.template
├── 45-roles.update
├── 49-autobind-services.update
├── 50-7_bit_check.update
├── 50-dogtag10-migration.update
├── 50-groupuuid.update
├── 50-hbacservice.update
├── 50-ipaconfig.update
├── 50-krbentypes.update
├── 50-nis.update
├── 55-pbacmemberof.update
├── 59-trusts-sysaccount.update
├── 60-trusts.update
├── 61-trusts-s4u2proxy.update
├── 62-ranges.update
├── 71-idviews-sasl-mapping.update
├── 71-idviews.update
├── 72-domainlevels.update
├── 73-certmap.update
├── 73-custodia.update
├── 73-passkey.update
├── 73-service-rbcd.update
├── 73-subid.update
├── 73-winsync.update
├── 75-user-trust-attributes.update
├── 80-schema_compat.update
├── 81-externalmembers.update
├── 90-post_upgrade_plugins.update
├── user_private_groups.ldif
├── uuid-conf.ldif
├── uuid.ldif
├── vault.ldif
├── version-conf.ldif
├── wsgi
│   └── plugins.py
├── wsgi.py
└── 20 directories, 253 files
[root@localhost ~]#
```

## Apache Web Server configuration:



```
[root@localhost ~]# cat /etc/httpd/conf.d/ipa.conf
#
# VERSION 34 - DO NOT REMOVE THIS LINE
# This file may be overwritten on upgrades.
#
# Load lookup_identity module in case it has not been loaded yet
# The module is used to search users according the certificate.
# IfModule !load_module modules/mod_lookup_identity.so
# IfModule !load_module modules/mod_lookup_identity.so
# IfModule !load_module modules/mod_lookup_identity.so

ProxyRequests Off

# We use chdir, a file format that the browser validates
DirectoryIndex index.html

# Substantially increase the request field size to support PG-PAC
# requests, ticket #2767. This should easily support a 64Kb PAC.
LimitRequestFieldSize 100000

# Increase connection keep alive time. Default value is 3 seconds, which is too
# short for interactive ipa commands. 30 seconds is a good compromise.
KeepAlive On
KeepAliveTimeout 30

# ipa-rewrite.conf is loaded separately

# Proper header for .tff fonts
AddType application/x-font-ttf ttf

# Enable compression
AddOutputFilterByType DEFLATE text/html text/plain text/xml \
  application/javascript application/json text/css \
  application/x-font-ttf
```

```
AbraLinux-Server  AbraLinux-Client
Activities  Terminal

[root@localhost ~]# cat /etc/httpd/conf.d/ssl.conf
#
# When we also provide SSL we have to listen to the
# standard HTTPS port in addition.
#
Listen 443 https

##
##  SSL Global Context
##
##  All SSL configuration in this context applies both to
##  the main server and all SSL-enabled virtual hosts.
##

#
#  Pass Phrase Dialog:
#  Configure the pass phrase gathering process.
#  The filtering dialog program ('builtin' is a internal
#  terminal dialog) has to provide the pass phrase on stdout.
SSLPassPhraseDialog exec:/usr/libexec/ipa/ipa-httpd-pwdreader

#
#  Inter-Process Session Cache:
#  Configure the SSL Session Cache: First the mechanism
#  to use and second the expiring timeout (in seconds).
SSLSessionCache         shmcb:/run/httpd/sslcach(512000)
SSLSessionCacheTimeout  300

#
#  Use "SSLCryptoDevice" to enable any supported hardware
#  accelerators. Use "openssl engine -v" to list supported
#  engine names. NOTE: If you enable an accelerator and the
#  server does not start, consult the error logs and ensure
#  your accelerator is functioning properly.
#
SSLCryptoDevice builtin
#SSLCryptoDevice ubsec

##
##  SSL Virtual Host Context
```

## Certificate Authority configuration:

```
AbraLinux-Server  AbraLinux-Client
Activities  Terminal
Jun 15 20:11
root@localhost:~

[root@localhost ~]# tree /etc/pki/pki-tomcat
/etc/pki/pki-tomcat
├── acme
│   ├── configsources.conf
│   ├── database.conf
│   ├── engine.conf
│   ├── issuer.conf
│   └── realm.conf
├── alias
│   ├── ca.crt
│   ├── cert9.db
│   ├── key4.db
│   ├── pkcs11.txt
│   └── pwdbfile.txt
├── ca
│   ├── adminCert.profile
│   ├── archives
│   │   ├── CS.cfg.bak.20260615195809
│   │   ├── CS.cfg.bak.20260615195923
│   │   ├── CS.cfg.bak.20260615195931
│   │   └── CS.cfg.bak.20260615195955
│   ├── caAuditSigningCert.profile
│   ├── caCert.profile
│   ├── caOCSPCert.profile
│   ├── CS.cfg
│   ├── CS.cfg.bak -> /var/lib/pki/pki-tomcat/conf/ca/archives/CS.cfg.bak.20260615195955
│   ├── CS.cfg.ipabkp
│   └── emails
│       ├── certIssued_CA
│       ├── certIssued_CA.html
│       ├── certIssued_RA
│       ├── certIssued_RA.html
│       ├── certRequestRejected.html
│       ├── certRevoked_CA
│       ├── certRevoked_CA.html
│       ├── certRevoked_RA
│       ├── certRevoked_RA.html
│       └── euJob1.html
```

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 15 20:12
root@localhost:~

├── caUserCert.cfg
├── caUserSMIMEcapCert.cfg
├── caUIDeviceCert.cfg
├── domainController.cfg
├── ECAdminCert.cfg
├── estServiceCert.cfg
├── proxy.conf
├── registry.cfg
├── serverCert.profile
├── subsystemCert.profile
├── Catalina
│   └── localhost
│       ├── acme.xml
│       ├── ca.xml
│       ├── pki.xml
│       ├── rewrite.config -> /usr/share/pki/server/conf/Catalina/localhost/rewrite.config
│       └── ROOT.xml
├── catalina.policy
├── catalina.properties -> /usr/share/pki/server/conf/catalina.properties
├── certs
│   ├── ca_admin.csr
│   ├── ca_audit_signing.csr
│   ├── ca_ocsp_signing.csr
│   ├── ca_signing.csr
│   ├── sslserver.csr
│   └── subsystem.csr
├── context.xml -> /etc/tomcat/context.xml
├── dogtag.keys
├── dogtag.keytab
├── logging.properties -> /usr/share/pki/server/conf/logging.properties
├── password.conf
├── serverCertNick.conf
├── server.xml
├── tomcat.conf
├── web.xml -> /etc/tomcat/web.xml
└── 10 directories, 170 files
root@localhost ~]#
```

Verify the installed IPA services:

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 15 20:13
root@localhost:~

[root@localhost ~]# systemctl status ipa
● ipa.service - Identity, Policy, Audit
   Loaded: loaded (/usr/lib/systemd/system/ipa.service; enabled; preset: disabled)
   Active: active (exited) since Mon 2026-06-15 20:01:35 EDT; 11min ago
   Process: 15703 ExecStart=/usr/sbin/ipactl start (code=exited, status=0/SUCCESS)
   Main PID: 15703 (code=exited, status=0/SUCCESS)
   CPU: 857ms

Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting Directory Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting krb5kdc Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting kadmind Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting named Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting httpd Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting ipa-custodia Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting pki-tomcatd Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting ipa-otpd Service
Jun 15 20:01:35 ipa.itmt15.local ipactl[15703]: Starting ipa-dnskeysyncd Service
Jun 15 20:01:35 ipa.itmt15.local systemd[1]: Finished Identity, Policy, Audit.
[root@localhost ~]#
```

Verify DNS:



```
Abraha-Server  [~] RHEL7-Client
Jun 15 20:18
root@localhost:~

[root@localhost ~]# kinit admin
Password for admin@ITMT15.LOCAL:
[root@localhost ~]# ipa dnszone-find
Zone name: itmt15.local.
Active zone: True
Authoritative nameserver: ipa.itmt15.local.
Administrator e-mail address: hostmaster.itmt15.local.
SOA serial: 1781568185
SOA refresh: 3600
SOA retry: 900
SOA expire: 1209600
SOA minimum: 3600
BIND update policy: grant ITMT15.LOCAL krb5-self * A; grant ITMT15.LOCAL krb5-self * AAAA; grant ITMT15.LOCAL krb5-self * SSHFP;
Dynamic update: True
Allow query: any;
Allow transfer: none;
-----
Number of entries returned 1
-----

[root@localhost ~]# ipa dnszone-add 15.168.192.in-addr.arpa --name-server=ipa.itmt15.local. --admin-email=admin@itmt15.local
ipa: WARNING: Semantic of setting Authoritative nameserver was changed. It is used only for setting the SOA MNAME attribute.
NS record(s) can be edited in zone apex - '@'.
Zone name: 15.168.192.in-addr.arpa.
Active zone: True
Authoritative nameserver: ipa.itmt15.local.
Administrator e-mail address: admin.itmt15.local
SOA serial: 1781569069
SOA refresh: 3600
SOA retry: 900
SOA expire: 1209600
SOA minimum: 3600
BIND update policy: grant ITMT15.LOCAL krb5-subdomain 15.168.192.in-addr.arpa. PTR;
Dynamic update: False
Allow query: any;
Allow transfer: none;
[root@localhost ~]# ipa dnszone-mod 15.168.192.in-addr.arpa. --dynamic-update=TRUE
Zone name: 15.168.192.in-addr.arpa.
Active zone: True
Authoritative nameserver: ipa.itmt15.local.
Administrator e-mail address: admin.itmt15.local
SOA serial: 1781569070
SOA refresh: 3600
SOA retry: 900
SOA expire: 1209600
SOA minimum: 3600
BIND update policy: grant ITMT15.LOCAL krb5-subdomain 15.168.192.in-addr.arpa. PTR;
Dynamic update: True
Allow query: any;
Allow transfer: none;
[root@localhost ~]#
```

Verify DNS again:

```
Abraha-Server  [~] RHEL7-Client
Jun 15 20:19
root@localhost:~

[root@localhost ~]# ipa dnszone-find
Zone name: 15.168.192.in-addr.arpa.
Active zone: True
Authoritative nameserver: ipa.itmt15.local.
Administrator e-mail address: admin.itmt15.local
SOA serial: 1781569070
SOA refresh: 3600
SOA retry: 900
SOA expire: 1209600
SOA minimum: 3600
BIND update policy: grant ITMT15.LOCAL krb5-subdomain 15.168.192.in-addr.arpa. PTR;
Dynamic update: True
Allow query: any;
Allow transfer: none;

Zone name: itmt15.local.
Active zone: True
Authoritative nameserver: ipa.itmt15.local.
Administrator e-mail address: hostmaster.itmt15.local.
SOA serial: 1781568185
SOA refresh: 3600
SOA retry: 900
SOA expire: 1209600
SOA minimum: 3600
BIND update policy: grant ITMT15.LOCAL krb5-self * A; grant ITMT15.LOCAL krb5-self * AAAA; grant ITMT15.LOCAL krb5-self * SSHFP;
Dynamic update: True
Allow query: any;
Allow transfer: none;
-----
Number of entries returned 2
-----
[root@localhost ~]#
```

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 15 20:20
root@localhost:~

[root@localhost ~]# nslookup ipa.itmt15.local
Server:      127.0.0.1
Address:     127.0.0.1#53

Name:   ipa.itmt15.local
Address: 192.168.15.10

[root@localhost ~]# dig ipa.itmt15.local

; <<>> DiG 9.16.23-RH <<>> ipa.itmt15.local
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 19002
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: 93e42905b9d30f3d010000006a3096fd64ea925fa7c4e5e3 (good)
;; QUESTION SECTION:
;ipa.itmt15.local.                IN      A

;; ANSWER SECTION:
ipa.itmt15.local.                1200    IN      A      192.168.15.10

;; Query time: 1 msec
;; SERVER: 127.0.0.1#53(127.0.0.1)
;; WHEN: Mon Jun 15 20:21:17 EDT 2026
;; MSG SIZE rcvd: 89
```

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 15 20:22
root@localhost:~

[root@localhost ~]# dig -x 192.168.15.10

; <<>> DiG 9.16.23-RH <<>> -x 192.168.15.10
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 11771
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 1232
;; COOKIE: e5638698a62cd9e8010000006a30975433a02c2034d0189e (good)
;; QUESTION SECTION:
;10.15.168.192.in-addr.arpa.      IN      PTR

;; AUTHORITY SECTION:
15.168.192.in-addr.arpa. 3600    IN      SOA     ipa.itmt15.local. admin.itmt15.local. 15.168.192.in-addr.arpa. 1781569070 3600 900 1209600 3600

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1)
;; WHEN: Mon Jun 15 20:22:44 EDT 2026
;; MSG SIZE rcvd: 177

[root@localhost ~]#
```

6. Add DNS records for the mail server (A, MX, PTR).

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 15 20:25
root@localhost:~

[root@localhost ~]# ipa dnsrecord-add itmt15.local mail --a-rec 192.168.15.10
Record name: mail
A record: 192.168.15.10
[root@localhost ~]# ipa dnsrecord-add itmt15.local @ --mx-rec="10 mail.itmt15.local."
Record name: @
MX record: 10 mail.itmt15.local.
NS record: ipa.itmt15.local.
[root@localhost ~]# ipa dnsrecord-add 15.168.192.in-addr.arpa 10 --ptr-rec=mail.itmt15.local.
Record name: 10
PTR record: mail.itmt15.local.
[root@localhost ~]#
```

## 7. Create users via command line and the Web UI.

### Command line:

```
AlmaLinux-Server x AlmaLinux-Client
Activities Terminal
[root@localhost ~]# kinit admin
Password for admin@ITMT15.LOCAL:
[root@localhost ~]# klist
Ticket cache: KCM0
Default principal: admin@ITMT15.LOCAL

Valid starting Expires Service principal
2026-06-15 20:26:21 2026-06-16 19:57:12 krbtgt/ITMT15.LOCAL@ITMT15.LOCAL
[root@localhost ~]# ipa user-find
-----
1 user matched
-----
User login: admin
Last name: Administrator
Home directory: /home/admin
Login shell: /bin/bash
Principal name: admin@ITMT15.LOCAL
Principal alias: admin@ITMT15.LOCAL, root@ITMT15.LOCAL
UID: 528600000
GID: 528600000
Account disabled: False
Number of entries returned 1

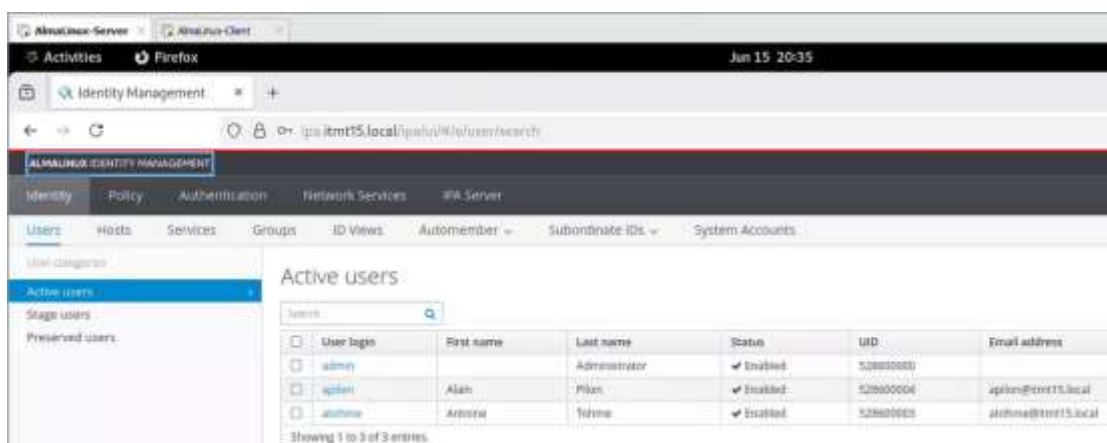
[root@localhost ~]# ipa user-add atohme --shell=/bin/bash --password
First name: Antoine
Last name: Tohme
Password:
Enter Password again to verify:
Added user "atohme"
-----
User login: atohme
First name: Antoine
Last name: Tohme
Full name: Antoine Tohme
Display name: Antoine Tohme
Initials: AT
Home directory: /home/atohme
GECOS: Antoine Tohme
Login shell: /bin/bash
Principal name: atohme@ITMT15.LOCAL
Principal alias: atohme@ITMT15.LOCAL
User password expiration: 20260616002755Z
Email address: atohme@itmt15.local
UID: 528600003
GID: 528600003
Password: True
Member of groups: ipausers
Kerberos keys available: True
[root@localhost ~]#
```

```
AlmaLinux-Server x AlmaLinux-Client
Activities Terminal
[root@localhost ~]# ipa user-find
-----
2 users matched
-----
User login: admin
Last name: Administrator
Home directory: /home/admin
Login shell: /bin/bash
Principal name: admin@ITMT15.LOCAL
Principal alias: admin@ITMT15.LOCAL, root@ITMT15.LOCAL
UID: 528600000
GID: 528600000
Account disabled: False

User login: atohme
First name: Antoine
Last name: Tohme
Home directory: /home/atohme
Login shell: /bin/bash
Principal name: atohme@ITMT15.LOCAL
Principal alias: atohme@ITMT15.LOCAL
Email address: atohme@itmt15.local
UID: 528600003
GID: 528600003
Account disabled: False
Number of entries returned 2
-----

[root@localhost ~]# ipa user-show atohme
User login: atohme
First name: Antoine
Last name: Tohme
Home directory: /home/atohme
Login shell: /bin/bash
Principal name: atohme@ITMT15.LOCAL
Principal alias: atohme@ITMT15.LOCAL
Email address: atohme@itmt15.local
UID: 528600003
GID: 528600003
Account disabled: False
Password: True
Member of groups: ipausers
Kerberos keys available: True
```

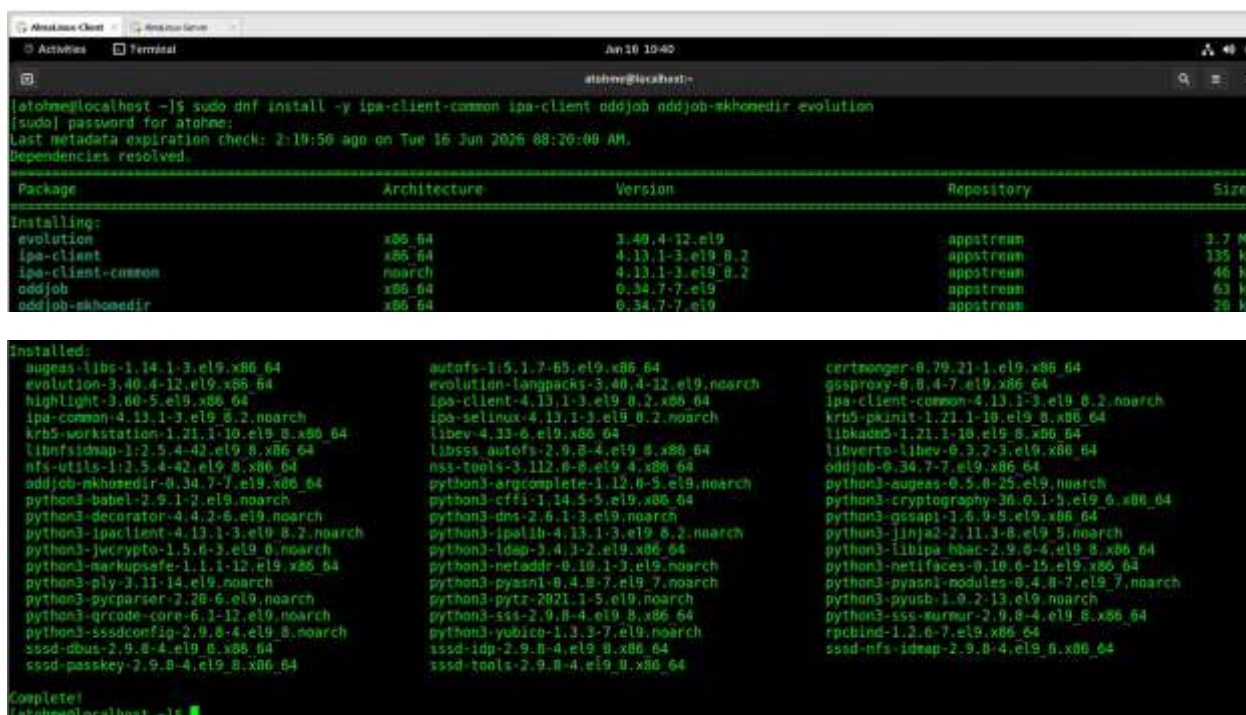
### Web UI:





## PART II - JOIN A LINUX CLIENT TO IPA DOMAIN

1. Log in **AlmaLinux-Client** graphically and install FreeIPA client and necessary packages.



```
atohme@localhost:~$ sudo dnf install -y ipa-client-common ipa-client oddjob oddjob-khomedir evolution
[sudo] password for atohme:
Last metadata expiration check: 2:19:56 ago on Tue 16 Jun 2026 08:20:00 AM.
Dependencies resolved.

```

| Package           | Architecture | Version          | Repository | Size  |
|-------------------|--------------|------------------|------------|-------|
| Installing:       |              |                  |            |       |
| evolution         | x86_64       | 3.40.4-12.el9    | appstream  | 3.7 M |
| ipa-client        | x86_64       | 4.13.1-3.el9_8.2 | appstream  | 135 k |
| ipa-client-common | noarch       | 4.13.1-3.el9_8.2 | appstream  | 46 k  |
| oddjob            | x86_64       | 0.34.7-7.el9     | appstream  | 63 k  |
| oddjob-khomedir   | x86_64       | 0.34.7-7.el9     | appstream  | 20 k  |

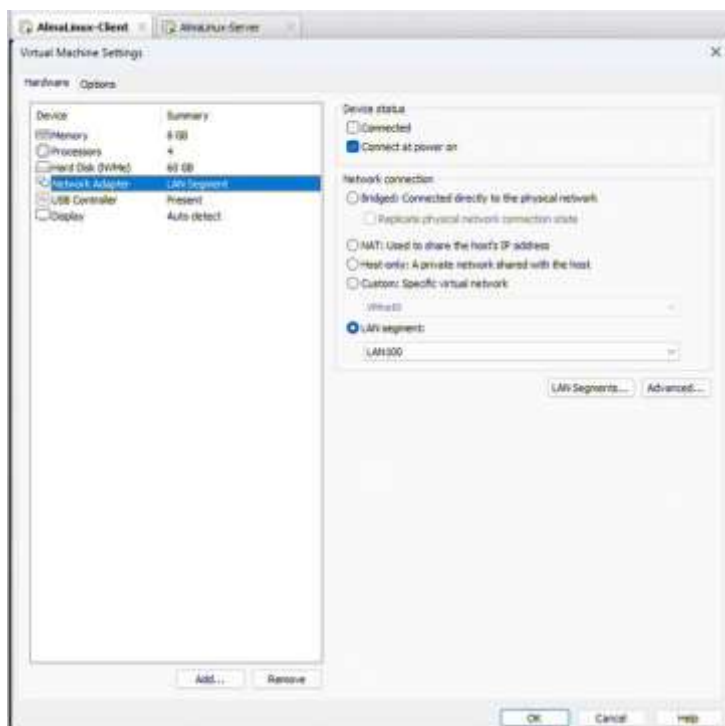
```

Installed:
augeas-libs-1.14.1-3.el9.x86_64
evolution-3.40.4-12.el9.x86_64
highlight-3.60-5.el9.x86_64
ipa-common-4.13.1-3.el9_8.2.noarch
krb5-workstation-1.21.1-10.el9_8.x86_64
libnfsidmap-1:2.5.4-42.el9_8.x86_64
nfs-utils-1:2.5.4-42.el9_8.x86_64
oddjob-khomedir-0.34.7-7.el9.x86_64
python3-babel-2.9.1-2.el9.noarch
python3-decorator-4.4.2-6.el9.noarch
python3-ipaclient-4.13.1-3.el9_8.2.noarch
python3-jwcrypto-1.5.6-3.el9_8.noarch
python3-markupsafe-1.1.1-12.el9.x86_64
python3-ply-3.11.14.el9.noarch
python3-pycompiler-2.70-6.el9.noarch
python3-pycode-core-6.1-12.el9.noarch
python3-sssdconfig-2.9.8-4.el9_8.noarch
sssd-dbus-2.9.8-4.el9_8.x86_64
sssd-passkey-2.9.8-4.el9_8.x86_64
autofs-1:5.1.7-65.el9.x86_64
evolution-langpacks-3.40.4-12.el9.noarch
ipa-client-4.13.1-3.el9_8.2.x86_64
ipa-selinux-4.13.1-3.el9_8.2.noarch
libev-4.33-6.el9.x86_64
libsss-autofs-2.9.8-4.el9_8.x86_64
nss-tools-3.112.6-8.el9_4.x86_64
python3-argcomplete-1.12.0-3.el9.noarch
python3-cffi-1.14.5-5.el9.x86_64
python3-dns-2.6.1-3.el9.noarch
python3-ipolih-4.13.1-3.el9_8.2.noarch
python3-ldap-3.4.3-2.el9.x86_64
python3-netaddr-0.10.1-3.el9.noarch
python3-pyasn1-0.4.8-7.el9_7.noarch
python3-pyts-2021.1-5.el9.noarch
python3-sss-2.9.8-4.el9_8.x86_64
python3-yubico-1.3.3-7.el9.noarch
sssd-idp-2.9.8-4.el9_8.x86_64
sssd-tools-2.9.8-4.el9_8.x86_64
certmonger-0.79.21-1.el9.x86_64
gssproxy-0.8.4-7.el9.x86_64
ipa-client-common-4.13.1-3.el9_8.2.noarch
krb5-pkinit-1.21.1-10.el9_8.x86_64
libkadm5-1.21.1-10.el9_8.x86_64
libverto-libev-0.3.2-3.el9.x86_64
oddjob-0.34.7-7.el9.x86_64
python3-augeas-0.5.0-25.el9.noarch
python3-cryptography-36.0.1-5.el9_6.x86_64
python3-gssapi-1.6.9-5.el9.x86_64
python3-jinja2-2.11.3-8.el9_5.noarch
python3-libipa-hbac-2.9.6-4.el9_8.x86_64
python3-netifaces-0.10.6-15.el9.x86_64
python3-pyasn1-modules-0.4.8-7.el9_7.noarch
python3-pyusb-1.0.2-13.el9.noarch
python3-sss-murmur-2.9.8-4.el9_8.x86_64
rpcbind-1.2.6-7.el9.x86_64
sssd-nfs-idmap-2.9.8-4.el9_8.x86_64

Complete!
atohme@localhost:~$
```

2. Configure client IP address, hostname, and /etc/hosts.

### LAN100



client IP address, hostname



```

[atohme@localhost ~]$ nmcli con del ethernet-ens160
Connection 'ethernet-ens160' (3e9d8b36-6b77-446e-bd76-e7eafae09403) successfully deleted.
[atohme@localhost ~]$ nmcli con add type ethernet ifname ens160 ipv4.method manual con-name LAN ipv4.addresses 192.168.15.200/24 ipv4.gateway 192.168.15.50 ipv4.dns 192.168.15.10
connection 'LAN' (80d14fee-650c-4f68-bc0b-580faf57fa95) successfully added.
[atohme@localhost ~]$ hostnamectl set-hostname client.itmt15.local
[atohme@localhost ~]$ ip address
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens160: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:0c:29:48:40:89 brd ff:ff:ff:ff:ff:ff
    altname enp3s0
    inet 192.168.15.200/24 brd 192.168.15.255 scope global noprefixroute ens160
        valid_lft forever preferred_lft forever
    inet6 fe80::1a30:dca1:28c2:d0b5/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[atohme@localhost ~]$ hostname
client.itmt15.local
[atohme@localhost ~]$

```

## /etc/hosts

```

127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1         localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.15.10 ipa.itmt15.local ipa mail.itmt15.local mai

```

```

[atohme@localhost ~]$ nslookup ipa.itmt15.local
Server:                192.168.15.10
Address:                192.168.15.10#53

Name:   ipa.itmt15.local
Address: 192.168.15.10

```

### 3. Join the client machine to the FreeIPA domain.

```

[atohme@localhost ~]$ sudo realm join --verbose itmt15.local
* Resolving: ldap.tcp.itmt15.local
* Performing LDAP DSE lookup on: 192.168.15.10
* Successfully discovered: itmt15.local

```

The screenshot shows the AlmaLinux Identity Management web interface in a Firefox browser. The URL is `ipa.itmt15.local/ui/#/e/host/search`. The page title is "ALMALINUX IDENTITY MANAGEMENT". The navigation bar includes tabs for Identity, Policy, Authentication, Network Services, and IPA Server. Under the IPA Server tab, there are sub-tabs: Users, Hosts (selected), Services, Groups, ID Views, Automember, Subordinate IDs, and System Accounts. The main content area is titled "Hosts" and contains a search bar. Below the search bar is a table with two columns: "Host name" and "Description". The table lists two entries: "client.itmt15.local" and "ipa.itmt15.local". At the bottom, it says "Showing 1 to 2 of 2 entries."

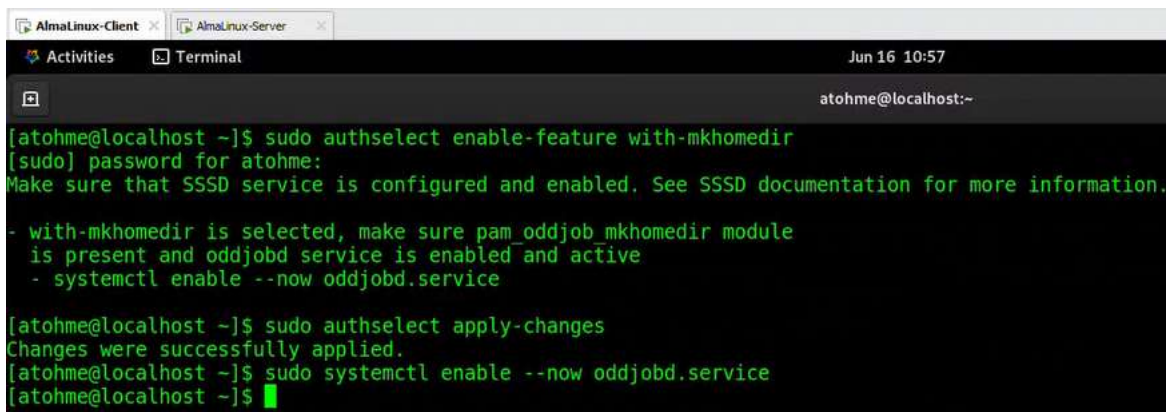
| Host name                                    | Description |
|----------------------------------------------|-------------|
| <input type="checkbox"/> client.itmt15.local |             |
| <input type="checkbox"/> ipa.itmt15.local    |             |

4. Create home and Maildir directories for test users.



```
AlmaLinux-Server AlmaLinux-Client
Jun 16 10:56
root@localhost:~
[root@localhost ~]# mkdir -p /home/apilon
[root@localhost ~]# chmod -R 700 /home/apilon
[root@localhost ~]# chown -R ailon:apilon /home/apilon
[root@localhost ~]# su - ailon -c "mkdir -p ~/Maildir/{cur,new,tmp,.Sent,.Trash,.Drafts,.Junk} && chmod -R 700 ~/Maildir"
[root@localhost ~]#
```

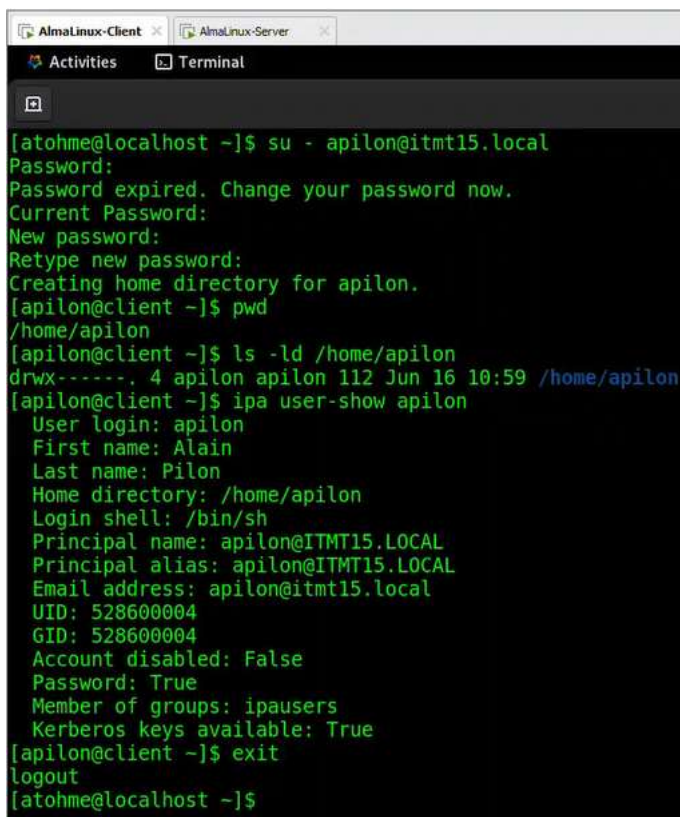
5. Enable pam\_mkhomedir and verify home folder creation.



```
AlmaLinux-Client AlmaLinux-Server
Jun 16 10:57
atohme@localhost:~
[atohme@localhost ~]$ sudo authselect enable-feature with-mkhomedir
[sudo] password for atohme:
Make sure that SSSD service is configured and enabled. See SSSD documentation for more information.
- with-mkhomedir is selected, make sure pam_oddjob_mkhomedir module
  is present and oddjobd service is enabled and active
- systemctl enable --now oddjobd.service

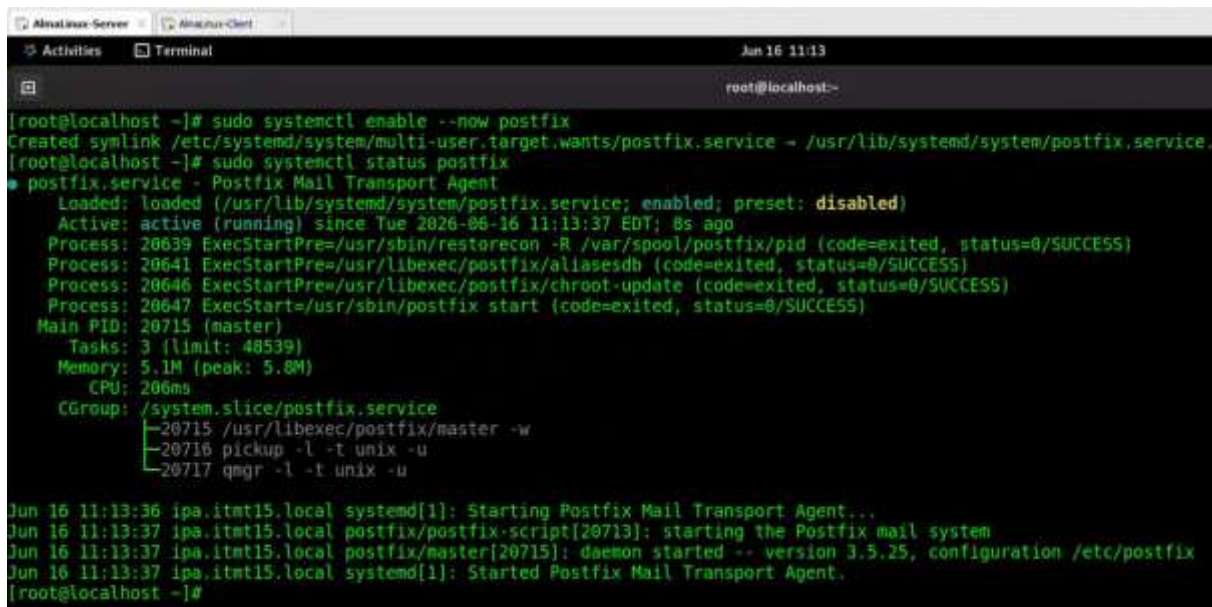
[atohme@localhost ~]$ sudo authselect apply-changes
Changes were successfully applied.
[atohme@localhost ~]$ sudo systemctl enable --now oddjobd.service
[atohme@localhost ~]$
```

6. Test user login and password change flow, then logout from the GUI session.



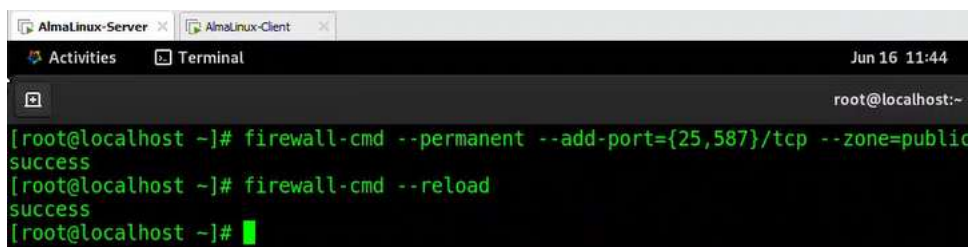
```
AlmaLinux-Client AlmaLinux-Server
[atohme@localhost ~]$ su - ailon@itmt15.local
Password:
Password expired. Change your password now.
Current Password:
New password:
Retype new password:
Creating home directory for ailon.
[ailon@client ~]$ pwd
/home/apilon
[ailon@client ~]$ ls -ld /home/apilon
drwx-----. 4 ailon ailon 112 Jun 16 10:59 /home/apilon
[ailon@client ~]$ ipa user-show ailon
User login: ailon
First name: Alain
Last name: Pilon
Home directory: /home/apilon
Login shell: /bin/sh
Principal name: ailon@ITMT15.LOCAL
Principal alias: ailon@ITMT15.LOCAL
Email address: ailon@itmt15.local
UID: 528600004
GID: 528600004
Account disabled: False
Password: True
Member of groups: ipausers
Kerberos keys available: True
[ailon@client ~]$ exit
logout
[atohme@localhost ~]$
```

1. On **AlmaLinux-Server** install Postfix and enable related firewall ports (25, 587).



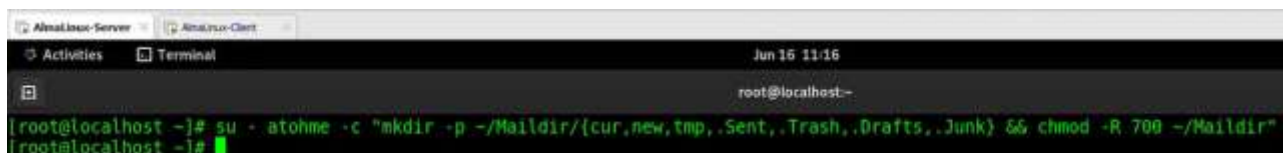
```
[root@localhost ~]# sudo systemctl enable --now postfix
Created symlink /etc/systemd/system/multi-user.target.wants/postfix.service - /usr/lib/systemd/system/postfix.service.
[root@localhost ~]# sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-06-16 11:13:37 EDT; 8s ago
     Process: 20639 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (code=exited, status=0/SUCCESS)
     Process: 20641 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)
     Process: 20646 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
     Process: 20647 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
   Main PID: 20715 (master)
     Tasks: 3 (limit: 48539)
    Memory: 5.1M (peak: 5.8M)
       CPU: 206ms
    CGroup: /system.slice/postfix.service
            └─20715 /usr/libexec/postfix/master -w
              └─20716 pickup -l -t unix -u
                └─20717 qmgr -l -t unix -u

Jun 16 11:13:36 ipa.itmt15.local systemd[1]: Starting Postfix Mail Transport Agent...
Jun 16 11:13:37 ipa.itmt15.local postfix/postfix-script[20713]: starting the Postfix mail system
Jun 16 11:13:37 ipa.itmt15.local postfix/master[20715]: daemon started -- version 3.5.25, configuration /etc/postfix
Jun 16 11:13:37 ipa.itmt15.local systemd[1]: Started Postfix Mail Transport Agent.
[root@localhost ~]#
```



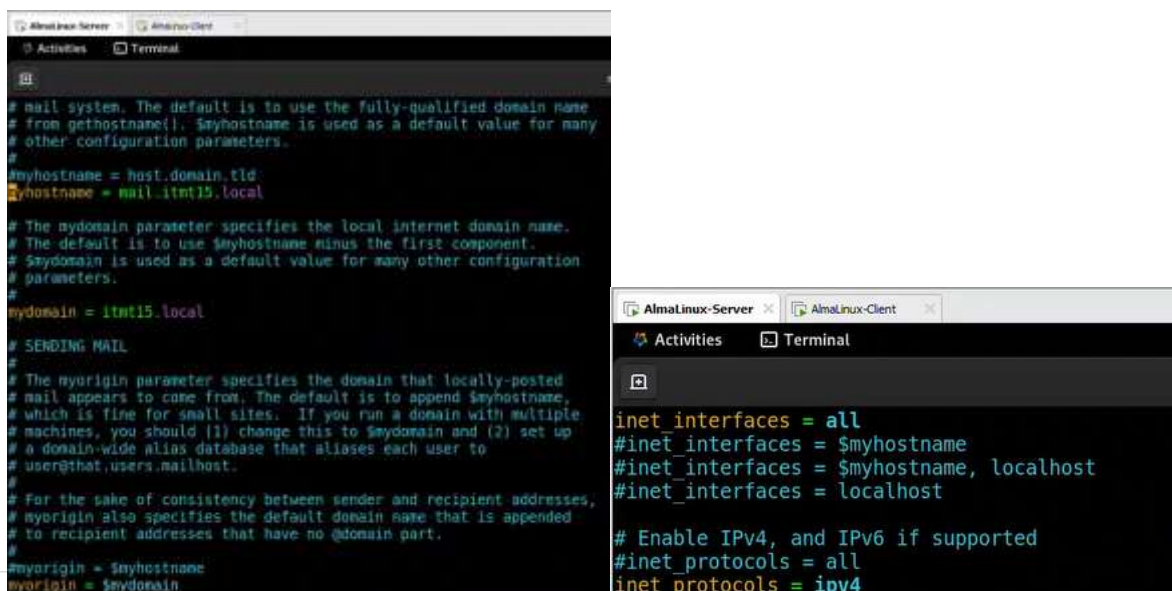
```
[root@localhost ~]# firewall-cmd --permanent --add-port={25,587}/tcp --zone=public
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]#
```

2. Create Maildir structure for users.



```
[root@localhost ~]# su - atohme -c "mkdir -p ~/Maildir/{cur,new,tmp,.Sent,.Trash,.Drafts,.Junk} && chmod -R 700 ~/Maildir"
[root@localhost ~]#
```

3. Modify main.cf with domain, hostname, SASL, and TLS settings.



```
# mail system. The default is to use the fully-qualified domain name
# from gethostname(). $myhostname is used as a default value for many
# other configuration parameters.
#
myhostname = host.domain.tld
myhostname = mail.itmt15.local

# The mydomain parameter specifies the local internet domain name.
# The default is to use $myhostname minus the first component.
# $mydomain is used as a default value for many other configuration
# parameters.
#
mydomain = itmt15.local

# SENDING MAIL
#
# The myorigin parameter specifies the domain that locally-posted
# mail appears to come from. The default is to append $myhostname,
# which is fine for small sites. If you run a domain with multiple
# machines, you should (1) change this to $mydomain and (2) set up
# a domain-wide alias database that aliases each user to
# user@that.users.mailhost.
#
# For the sake of consistency between sender and recipient addresses,
# myorigin also specifies the default domain name that is appended
# to recipient addresses that have no @domain part.
#
myorigin = $myhostname
myorigin = $mydomain
```

```
inet_interfaces = all
inet_interfaces = $myhostname
inet_interfaces = $myhostname, localhost
inet_interfaces = localhost

# Enable IPv4, and IPv6 if supported
inet_protocols = all
inet_protocols = ipv4
```



```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal
root

mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
#mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain,
# mail.$mydomain, www.$mydomain, ftp.$mydomain

# Enable SASL
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth

# Require authentication for clients
smtpd_recipient_restrictions =
    permit_sasl_authenticated,
    permit_mynetworks,
    reject_unauth_destination

# Optional: Secure connection
smtpd_tls_auth_only = yes
"/etc/postfix/main.cf" 752L, 29699B

#home_mailbox = Mailbox
home_mailbox = Maildir/
```

4. Edit master.cf to enable submission service.

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal

#dnsblog unix - - n - 0 dnsblog
#tlsproxy unix - - n - 0 tlsproxy
submission inet n - n - - smtpd
-o syslog_name=postfix/submission
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
-o smtpd_tls_auth_only=yes
# -o smtpd_reject_unlisted_recipient=no
# -o smtpd_client_restrictions=$mua_client_restrictions
# -o smtpd_helo_restrictions=$mua_helo_restrictions
# -o smtpd_sender_restrictions=$mua_sender_restrictions
-o smtpd_recipient_restrictions=permit_sasl_authenticated,reject
-o smtpd_relay_restrictions=permit_sasl_authenticated,reject
```

5. Start Postfix service and verify its status.

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal
Jun 16 11:40
root@localhost:~

[root@localhost ~]# sudo systemctl restart postfix
[root@localhost ~]# sudo systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-06-16 11:40:30 EDT; 4s ago
     Process: 20966 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (code=exited, status=0/SUCCESS)
     Process: 20967 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)
     Process: 20969 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
     Process: 20970 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
  Main PID: 21039 (master)
    Tasks: 3 (limit: 48539)
   Memory: 4.4M (peak: 5.3M)
      CPU: 199ms
   CGroup: /system.slice/postfix.service
           └─21039 /usr/libexec/postfix/master -w
             └─21040 pickup -l -t unix -u
               └─21041 qmgr -l -t unix -u

Jun 16 11:40:30 ipa.itmt15.local systemd[1]: Starting Postfix Mail Transport Agent...
Jun 16 11:40:30 ipa.itmt15.local postfix/master[21039]: daemon started -- version 3.5.25, configuration /etc/postfix
Jun 16 11:40:30 ipa.itmt15.local systemd[1]: Started Postfix Mail Transport Agent.
[root@localhost ~]#
```



```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 16 11:42
root@localhost:~

[root@localhost ~]# postconf -n
alias database = hash:/etc/aliases
alias maps = hash:/etc/aliases
command_directory = /usr/sbin
compatibility_level = 2
daemon_directory = /usr/libexec/postfix
data_directory = /var/lib/postfix
debug_peer_level = 2
debugger_command = PATH=/bin:/usr/bin:/usr/local/bin:/usr/X11R6/bin ddd $daemon_directory/$process_name $process_id & sleep 5
home_mailbox = Maildir/
html_directory = no
inet_interfaces = all
inet_protocols = ipv4
mail_owner = postfix
mailq_path = /usr/bin/mailq.postfix
manpage_directory = /usr/share/man
meta_directory = /etc/postfix
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mydomain = itmt15.local
myhostname = mail.itmt15.local
myorigin = $mydomain
newaliases_path = /usr/bin/newaliases.postfix
queue_directory = /var/spool/postfix
readme_directory = /usr/share/doc/postfix/README_FILES
sample_directory = /usr/share/doc/postfix/samples
sendmail_path = /usr/sbin/sendmail.postfix
setgid_group = postdrop
shlib_directory = /usr/lib64/postfix
smtp_tls_CAfile = /etc/pki/tls/certs/ca-bundle.crt
smtp_tls_CAspath = /etc/pki/tls/certs
smtp_tls_security_level = may
smtpd_recipient_restrictions = permit_sasl_authenticated, permit_mynetworks, reject_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_path = private/auth
smtpd_sasl_type = dovecot
smtpd_tls_auth_only = yes
smtpd_tls_cert_file = /etc/pki/tls/certs/postfix.pem
smtpd_tls_key_file = /etc/pki/tls/private/postfix.key
```

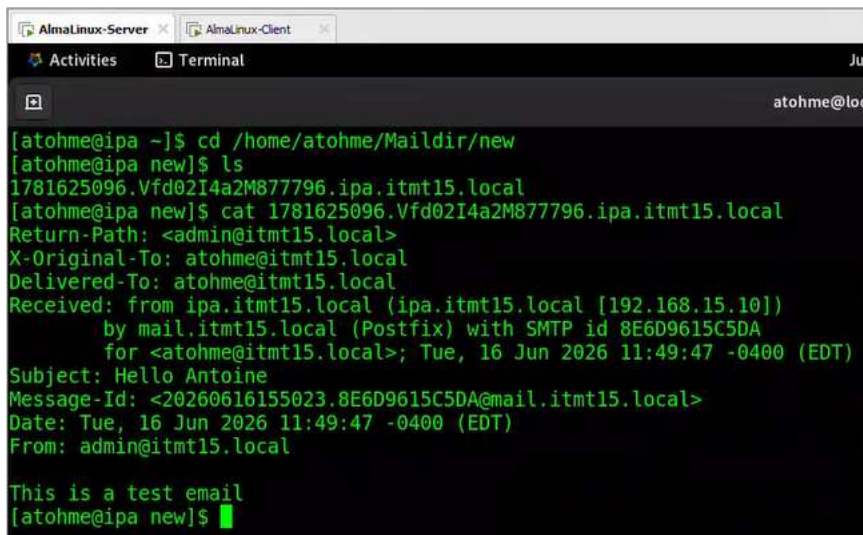
6. Send a test email using telnet.

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal
[root@ipa ~]# telnet mail.itmt15.local 25
Trying 192.168.15.10...
Connected to mail.itmt15.local.
Escape character is '^]'.
220 mail.itmt15.local ESMTP Postfix
mail from:admin@itmt15.local
250 2.1.0 Ok
rcpt to:atohme@itmt15.local
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Subject: Hello Antoine
This is a test email
.
250 2.0.0 Ok: queued as 8E6D9615C5DA
quit
221 2.0.0 Bye
Connection closed by foreign host.
[root@ipa ~]#
```

7. Check Postfix logs and user Maildir to confirm delivery.

```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 16 11:53
root@localhost:~

[root@localhost ~]# tail -f /var/log/maillog
Jun 16 11:37:52 localhost postfix[18998]: fatal: /etc/postfix/main.cf, line 747: missing '=' after attribute name: "permit_sasl_authenticated,"
Jun 16 11:40:30 localhost postfix/postfix-script[21837]: starting the Postfix mail system
Jun 16 11:40:30 localhost postfix/master[21839]: daemon started -- version 3.5.25, configuration /etc/postfix
Jun 16 11:48:35 localhost postfix/smtpd[21265]: connect from ipa.itmt15.local[192.168.15.10]
Jun 16 11:50:23 localhost postfix/smtpd[21265]: 0E6D9615C5DA: client=ipa.itmt15.local[192.168.15.10]
Jun 16 11:51:36 localhost postfix/cleanup[21280]: 0E6D9615C5DA: message-id=<20260616155023.0E6D9615C5DA@mail.itmt15.local>
Jun 16 11:51:36 localhost postfix/qmgr[21041]: 0E6D9615C5DA: from=admin@itmt15.local, size=374, nrcpt=1 (queue active)
Jun 16 11:51:36 localhost postfix/local[21286]: 0E6D9615C5DA: to=atohme@itmt15.local, relay=local, delay=109, delays=109/0.01/0/0, dsn=2.0.0, status=sent (delivered to maildir)
Jun 16 11:51:36 localhost postfix/qmgr[21041]: 0E6D9615C5DA: removed
Jun 16 11:51:52 localhost postfix/smtpd[21265]: disconnect from ipa.itmt15.local[192.168.15.10] mail=1 rcpt=1 data=1 quit=1 commands=4
```

A terminal window titled 'AlmaLinux-Server' and 'AlmaLinux-Client' showing a user 'atohme@ipa' navigating to a directory and displaying the contents of an email file. The email is from 'admin@itmt15.local' to 'atohme@itmt15.local' with the subject 'Hello Antoine'.

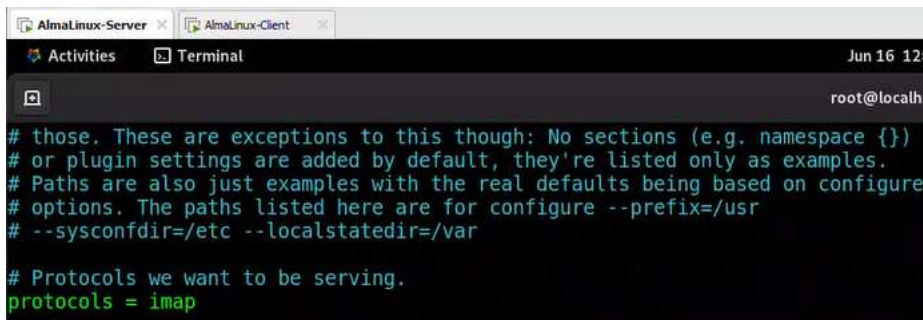
```
[atohme@ipa ~]$ cd /home/atohme/Maildir/new
[atohme@ipa new]$ ls
1781625096.Vfd02I4a2M877796.ipa.itmt15.local
[atohme@ipa new]$ cat 1781625096.Vfd02I4a2M877796.ipa.itmt15.local
Return-Path: <admin@itmt15.local>
X-Original-To: atohme@itmt15.local
Delivered-To: atohme@itmt15.local
Received: from ipa.itmt15.local (ipa.itmt15.local [192.168.15.10])
        by mail.itmt15.local (Postfix) with SMTP id 8E6D9615C5DA
        for <atohme@itmt15.local>; Tue, 16 Jun 2026 11:49:47 -0400 (EDT)
Subject: Hello Antoine
Message-Id: <20260616155023.8E6D9615C5DA@mail.itmt15.local>
Date: Tue, 16 Jun 2026 11:49:47 -0400 (EDT)
From: admin@itmt15.local

This is a test email
[atohme@ipa new]$
```

## PART IV - DOVECOT IMAP SERVER CONFIGURATION

1. On **AlmaLinux-Server** Install Dovecot.
2. Configure dovecot.conf, 10-mail.conf, 10-master.conf, and 10-auth.conf with IMAP, Maildir, and SASL settings.

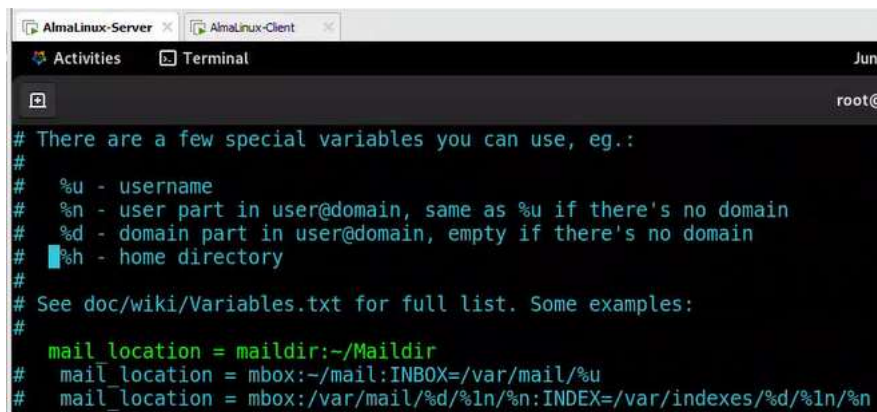
### dovecot.conf

A terminal window showing the configuration of dovecot.conf. The user is root@localhost and is editing the file. The configuration sets protocols to imap.

```
# those. These are exceptions to this though: No sections (e.g. namespace {})
# or plugin settings are added by default, they're listed only as examples.
# Paths are also just examples with the real defaults being based on configure
# options. The paths listed here are for configure --prefix=/usr
# --sysconfdir=/etc --localstatedir=/var

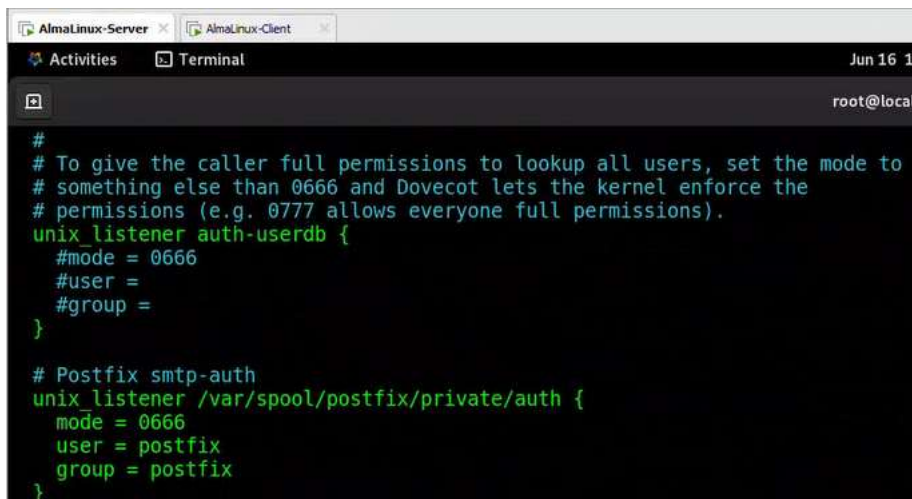
# Protocols we want to be serving.
protocols = imap
```

### 10-mail.conf

A terminal window showing the configuration of 10-mail.conf. The user is root@localhost and is editing the file. The configuration sets mail\_location to maildir:~/Maildir.

```
# There are a few special variables you can use, eg.:
#
# %u - username
# %n - user part in user@domain, same as %u if there's no domain
# %d - domain part in user@domain, empty if there's no domain
# %h - home directory
#
# See doc/wiki/Variables.txt for full list. Some examples:
#
mail_location = maildir:~/Maildir
# mail_location = mbox:~/mail:INBOX=/var/mail/%u
# mail_location = mbox:/var/mail/%d/%n/INDEX=/var/indexes/%d/%n/%n
```

### 10-master.conf

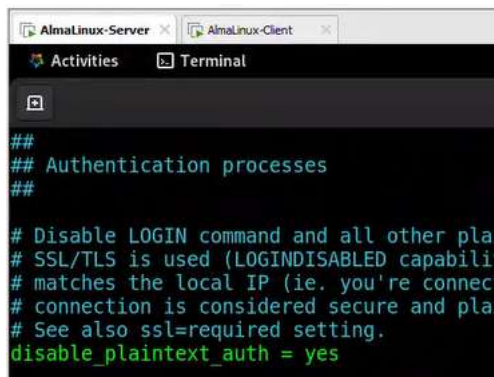


```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 16 1
root@localh

#
# To give the caller full permissions to lookup all users, set the mode to
# something else than 0666 and Dovecot lets the kernel enforce the
# permissions (e.g. 0777 allows everyone full permissions).
unix_listener auth-userdb {
    #mode = 0666
    #user =
    #group =
}

# Postfix smtp-auth
unix_listener /var/spool/postfix/private/auth {
    mode = 0666
    user = postfix
    group = postfix
}
```

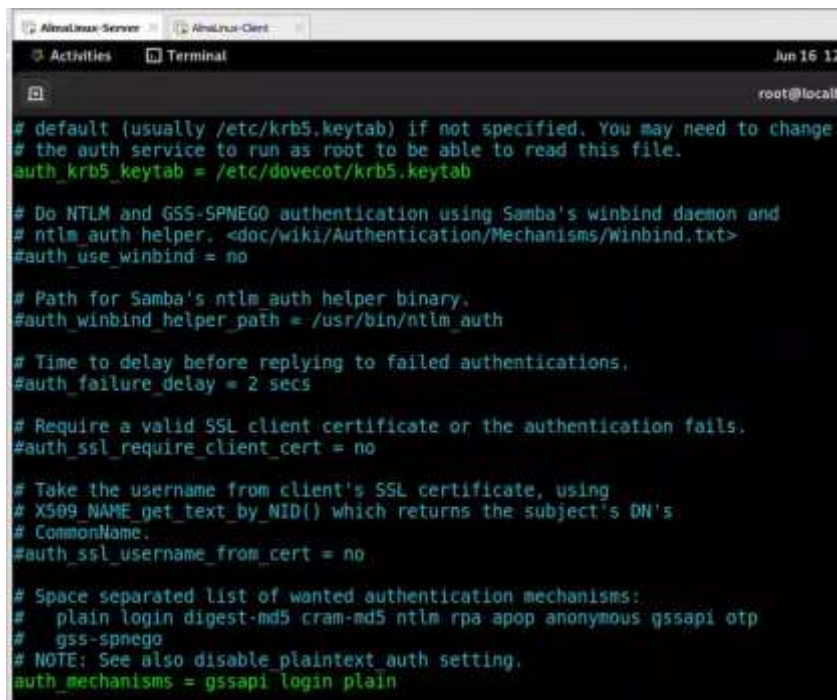
## 10-auth.conf



```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal
root@localh

##
## Authentication processes
##

# Disable LOGIN command and all other plaintext authentication
# SSL/TLS is used (LOGINDISABLED capability)
# matches the local IP (ie. you're connecting to localhost)
# connection is considered secure and plaintext authentication is
# See also ssl=required setting.
disable_plaintext_auth = yes
```



```
AlmaLinux-Server AlmaLinux-Client
Activities Terminal Jun 16 12
root@localh

# default (usually /etc/krb5.keytab) if not specified. You may need to change
# the auth service to run as root to be able to read this file.
auth_krb5_keytab = /etc/dovecot/krb5.keytab

# Do NTLM and GSS-SPNEGO authentication using Samba's winbind daemon and
# ntlm_auth helper. <doc/wiki/Authentication/Mechanisms/Winbind.txt>
#auth_use_winbind = no

# Path for Samba's ntlm_auth helper binary.
#auth_winbind_helper_path = /usr/bin/ntlm_auth

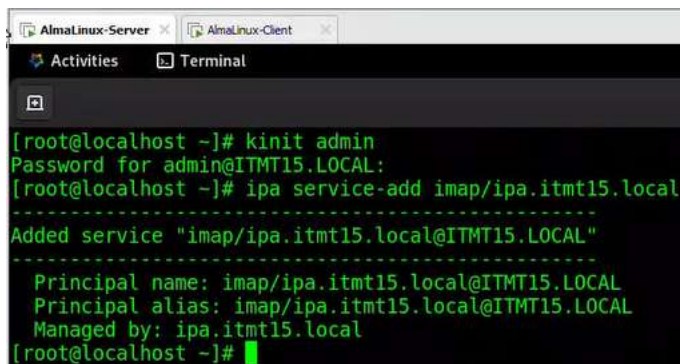
# Time to delay before replying to failed authentications.
#auth_failure_delay = 2 secs

# Require a valid SSL client certificate or the authentication fails.
#auth_ssl_require_client_cert = no

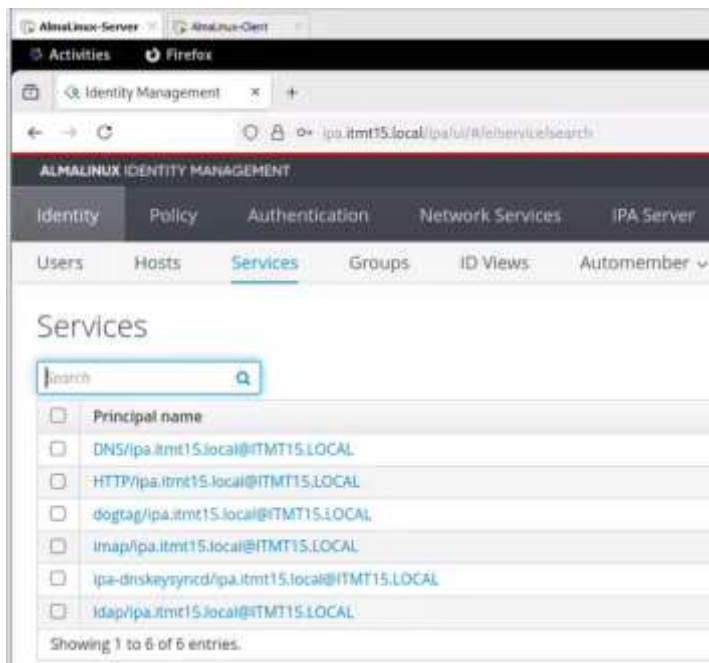
# Take the username from client's SSL certificate, using
# X509_NAME_get_text_by_NID() which returns the subject's DN's
# CommonName.
#auth_ssl_username_from_cert = no

# Space separated list of wanted authentication mechanisms:
#   plain login digest-md5 cram-md5 ntlm rpa apop anonymous gssapi otp
#   gss-spnego
# NOTE: See also disable_plaintext_auth setting.
auth_mechanisms = gssapi login plain
```

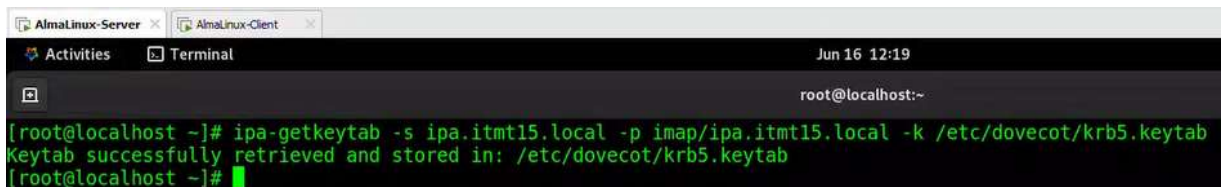
3. Add and verify an IMAP Kerberos service in FreeIPA.



```
[root@localhost ~]# kinit admin
Password for admin@ITMT15.LOCAL:
[root@localhost ~]# ipa service-add imap/ipa.itmt15.local
-----
Added service "imap/ipa.itmt15.local@ITMT15.LOCAL"
-----
Principal name: imap/ipa.itmt15.local@ITMT15.LOCAL
Principal alias: imap/ipa.itmt15.local@ITMT15.LOCAL
Managed by: ipa.itmt15.local
[root@localhost ~]#
```

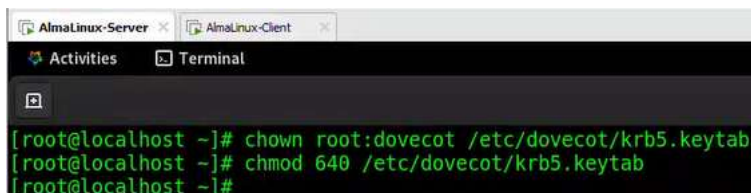


4. Generate and install the Kerberos keytab.



```
[root@localhost ~]# ipa-getkeytab -s ipa.itmt15.local -p imap/ipa.itmt15.local -k /etc/dovecot/krb5.keytab
Keytab successfully retrieved and stored in: /etc/dovecot/krb5.keytab
[root@localhost ~]#
```

5. Set secure permissions on keytab file.



```
[root@localhost ~]# chown root:dovecot /etc/dovecot/krb5.keytab
[root@localhost ~]# chmod 640 /etc/dovecot/krb5.keytab
[root@localhost ~]#
```

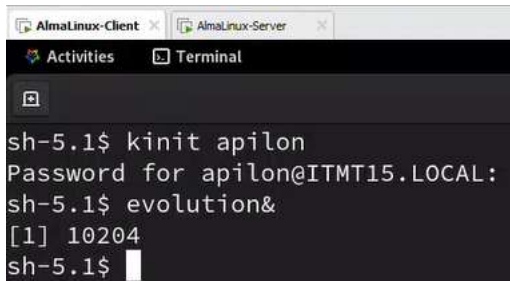
6. Start Dovecot service and open firewall ports (143, 993).





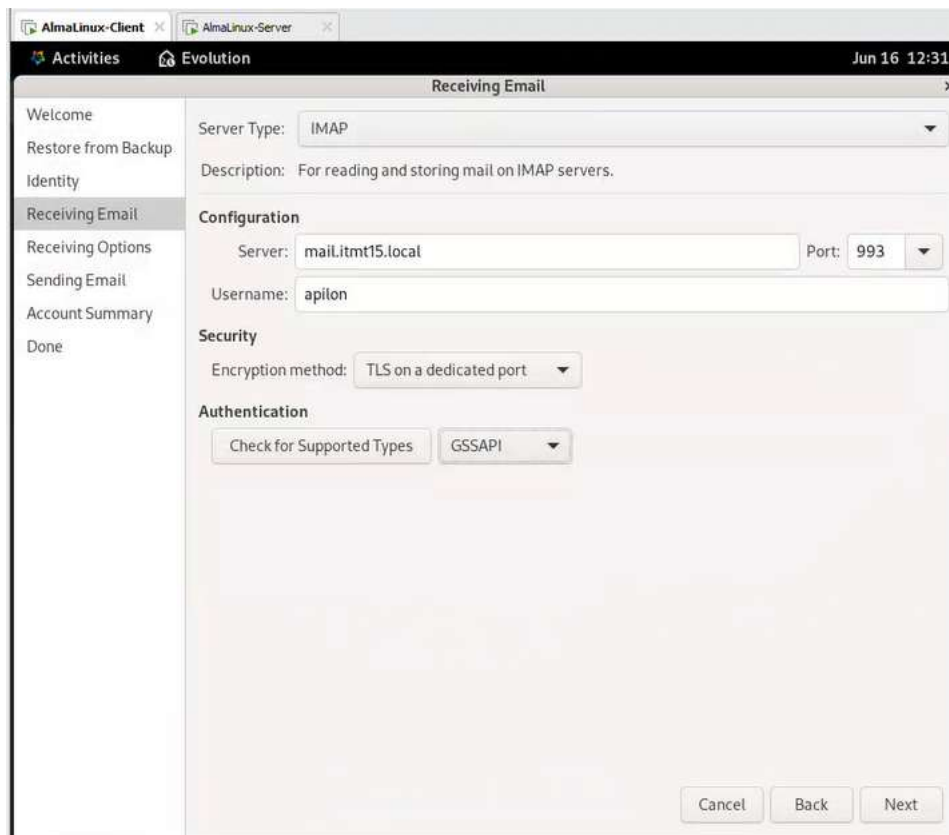
## PART V - EMAIL CLIENT (EVOLUTION) CONFIGURATION

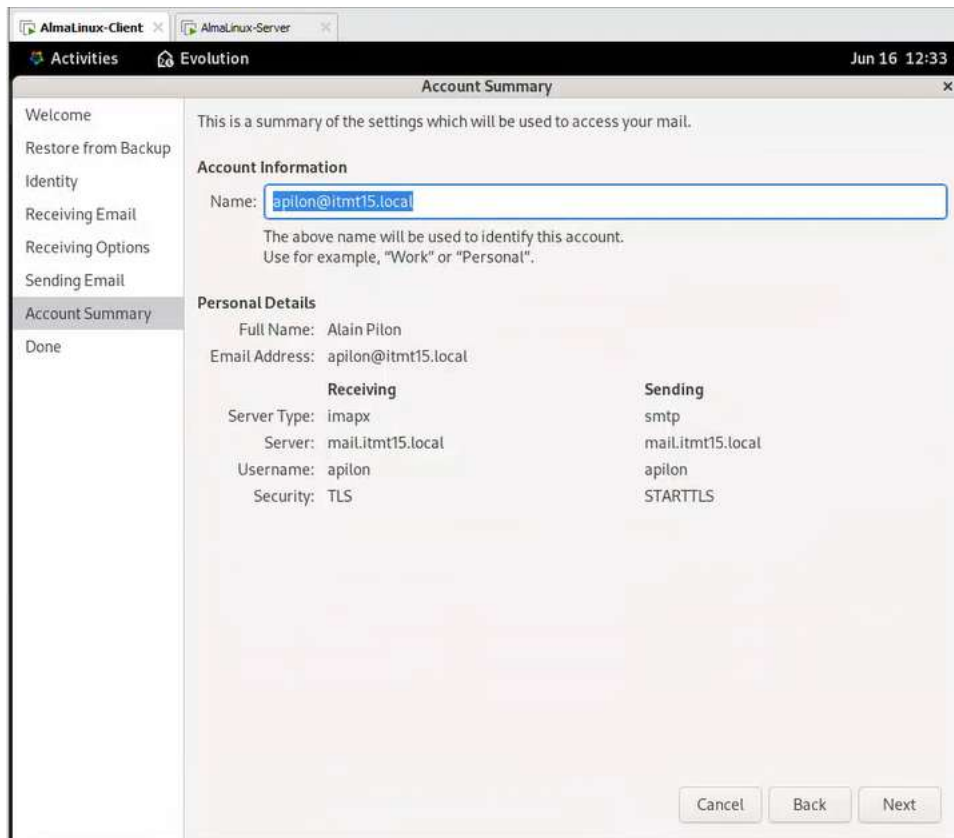
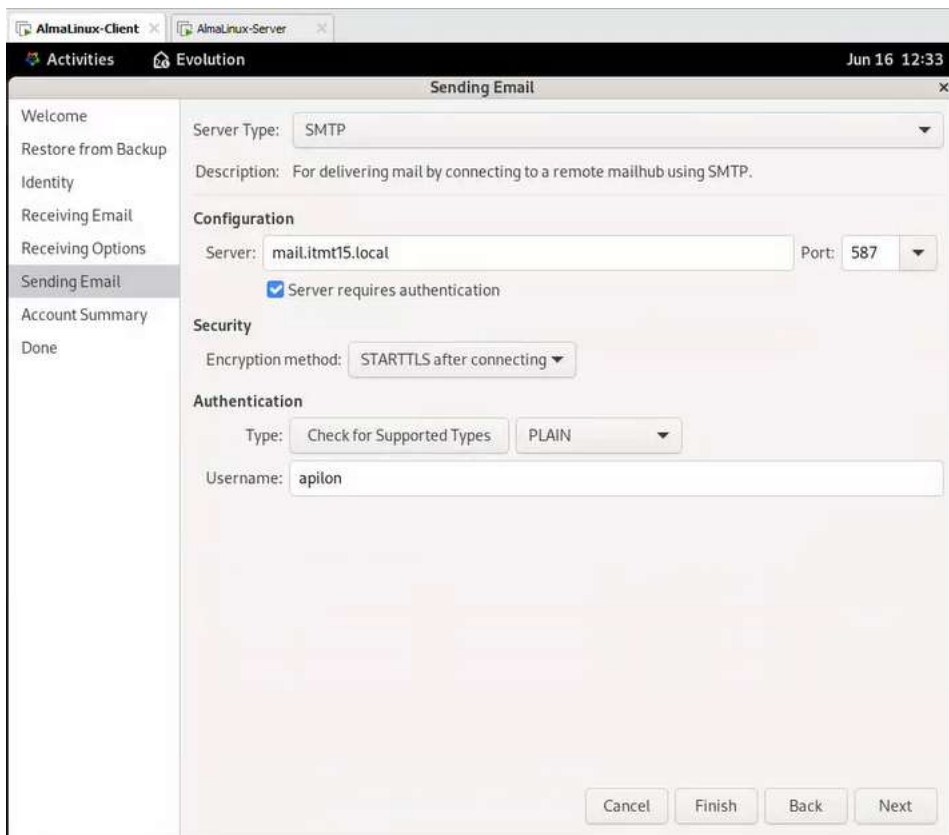
1. Log in **AlmaLinux-Client** graphically with domain user **apilon@itmtXX.local**.
2. Install **Evolution** email client.
3. Use kinit to authenticate the user.



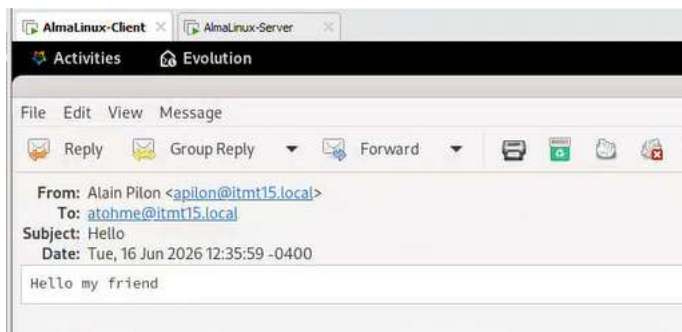
```
sh-5.1$ kinit apilon
Password for apilon@ITMT15.LOCAL:
sh-5.1$ evolution&
[1] 10204
sh-5.1$
```

4. Configure Evolution for IMAP (TLS) and SMTP (STARTTLS) using Kerberos.





5. Send a test email to another user.



## 6. Verify email delivery on the server and via client.

```

[root@ipa ~]# tail -f /var/log/maillog
Jun 16 12:35:59 localhost postfix/submission/smtpd[21745]: connect from unknown[192.168.15.200]
Jun 16 12:37:13 localhost postfix/submission/smtpd[21745]: 17E36615C5F7: client=unknown[192.168.15.200], sasl_method=PLAIN, sasl_username=apilon
Jun 16 12:37:13 localhost postfix/cleanup[21763]: 17E36615C5F7: message-id=<f71c4276fec094470c95f3e6b5d56dbe772e3248.camel@itmt15.local>
Jun 16 12:37:13 localhost postfix/qmgr[21841]: 17E36615C5F7: from=apilon@itmt15.local, size=524, nrcpt=1 (queue active)
Jun 16 12:37:13 localhost postfix/submission/smtpd[21745]: disconnect from unknown[192.168.15.200] ehlo=2 starttls=1 auth=1 mail=1 rcpt=1 data=1 quit=1
commands=4
Jun 16 12:37:13 localhost postfix/local[21764]: 17E36615C5F7: to=atohme@itmt15.local, relay=local, delay=0.00, delays=0.05/0.01/0/0, dsn=2.0.0, stat
us=sent (delivered to maildir)
Jun 16 12:37:13 localhost postfix/qmgr[21841]: 17E36615C5F7: removed
Jun 16 12:40:33 localhost postfix/anvil[21747]: statistics: max connection rate 1/60s for (submission:192.168.15.200) at Jun 16 12:35:59
Jun 16 12:40:33 localhost postfix/anvil[21747]: statistics: max connection count 1 for (submission:192.168.15.200) at Jun 16 12:35:59
Jun 16 12:40:33 localhost postfix/anvil[21747]: statistics: max cache size 1 at Jun 16 12:35:59

```

```

[root@localhost ~]# sudo journalctl -u dovecot -f
Jun 16 12:20:46 ipa.itmt15.local systemd[1]: Starting Dovecot IMAP/POP3 email server...
Jun 16 12:20:47 ipa.itmt15.local dovecot[21640]: master: Dovecot v2.3.16 (7c2e990c1a) starting up for imap
Jun 16 12:20:47 ipa.itmt15.local systemd[1]: Started Dovecot IMAP/POP3 email server.
Jun 16 12:31:11 ipa.itmt15.local dovecot[21650]: imap-login: Disconnected: Connection closed: read(size=503) failed: Connection reset by peer (no auth
attempts in 2 secs): user=*, rip=192.168.15.200, lip=192.168.15.10, TLS Handshaking: read(size=503) failed: Connection reset by peer, session=LhsPW
GFUSpZAgA/I>
Jun 16 12:34:35 ipa.itmt15.local dovecot[21650]: imap-login: Login: user=apilon, method=GSSAPI, rip=192.168.15.200, lip=192.168.15.10, mpid=21736, T
LS, session=zm0HGFU/oPAgA/I>
Jun 16 12:34:35 ipa.itmt15.local dovecot[21650]: imap-login: Login: user=apilon, method=GSSAPI, rip=192.168.15.200, lip=192.168.15.10, mpid=21739, T
LS, session=zm0HGFU/oPAgA/I>
Jun 16 12:37:12 ipa.itmt15.local auth[21752]: pam_unix(dovecot-auth): authentication success; logname=uid=0 euid=0 tty=dovecot ruser=apilon rhost=192.
168.15.200 user=apilon

```

```

[atohme@ipa ~]$ tree Maildir/
Maildir/
├── cur
├── new
│   ├── 1781625096.Vfd02I4a2M877796.ipa.itmt15.local
│   └── 1781627833.Vfd02I4b6M148413.ipa.itmt15.local
└── tmp

3 directories, 2 files
[atohme@ipa ~]$ cat Maildir/new/1781627833.Vfd02I4b6M148413.ipa.itmt15.local
Return-Path: <apilon@itmt15.local>
X-Original-To: atohme@itmt15.local
Delivered-To: atohme@itmt15.local
Received: from client.itmt15.local (unknown [192.168.15.200])
        by mail.itmt15.local (Postfix) with ESMTPSA id 17E36615C5F7
        for <atohme@itmt15.local>; Tue, 16 Jun 2026 12:37:13 -0400 (EDT)
Message-ID: <f71c4276fec094470c95f3e6b5d56dbe772e3248.camel@itmt15.local>
Subject: Hello
From: Alain Pilon <apilon@itmt15.local>
To: atohme@itmt15.local
Date: Tue, 16 Jun 2026 12:35:59 -0400
Content-Type: text/plain
User-Agent: Evolution 3.40.4 (3.40.4-12.el9)
MIME-Version: 1.0
Content-Transfer-Encoding: 7bit

Hello my friend

[atohme@ipa ~]$

```



